

Week 11 — Diagnostics, Model Selection & Long Memory

7 May 2026

The big picture. Validate a model through its residuals (Ljung–Box), select via AIC/AICc/BIC (Box–Jenkins); long memory describes slowly decaying autocovariances.

Once an ARIMA model is fitted, two questions remain: *does it fit?* (diagnostics, from the residuals) and *is it best among candidates?* (selection, from information criteria). We then leave the ARMA world: ARMA covariances decay *exponentially*, but many real series decay only *polynomially* — this is **long memory**.

Key definitions

Definition — Residuals

For a fitted model with one-step fitted value $\hat{X}_t$, the **residuals** and **standardized residuals** are

$$e_t = X_t - \hat{X}_t, \quad \tilde{e}_t = \frac{e_t}{\sqrt{\text{Var}(e_t)}}.$$

If the model is right, the e_t should behave like the driving white noise. Four checks: constant zero **mean**, constant **variance** (homoscedasticity), **uncorrelated** in t , and **Gaussian**.

Why residuals are the test

A correct model has captured all the linearly predictable structure, so what is left over, $e_t = X_t - \hat{X}_t$, should be the unpredictable innovation ε_t . Every property we required of the white-noise input is exactly what we now test in the residuals. Each failure points at one flaw: a trend left in (mean), volatility clustering (variance), an unmodelled AR/MA term (correlation), or non-normal innovations (Gaussianity).

Three checks are visual: plot $\{(t, \tilde{e}_t)\}$ for the mean (no trend, hover at 0) and variance (constant spread, no fanning); a **Q–Q plot** vs. normal quantiles checks (marginal) Gaussianity — straight line if Gaussian, curling tails if heavy-tailed. The fourth is quantitative.

Definition — Residual correlogram & its band

The residual sample autocorrelation at lag τ is

$$\hat{r}_\tau = \frac{\sum_{t=1}^{n-\tau} (e_{t+\tau} - \bar{e})(e_t - \bar{e})}{\sum_{t=1}^n (e_t - \bar{e})^2}.$$

Under whiteness each $\hat{r}_\tau \approx \mathcal{N}(0, 1/n)$, so compare against the pointwise 95% band $(-1.96/\sqrt{n}, 1.96/\sqrt{n})$.

Definition — Information criteria: AIC, AICc, BIC

For k estimated parameters, n observations, maximized likelihood L :

$$\begin{aligned} \text{AIC} &= -2 \log L + 2k, \\ \text{AICc} &= -2 \log L + 2k \frac{n}{n-k-1} = \text{AIC} + \frac{2k(k+1)}{n-k-1}, \\ \text{BIC} &= -2 \log L + k \log n. \end{aligned}$$

Smaller is better. For ARMA(p, q), $k = p + q + 1$ (p AR, q MA, plus σ^2). AIC tends to

over-estimate p ; AICc corrects the small-sample bias (and $\rightarrow$ AIC as $n \rightarrow \infty$); BIC penalizes more (for $n > e^2$) and favours parsimony.

Definition — Long memory (covariance & spectrum)

A stationary process has **long memory** if its ACVS decays only polynomially,

$$\gamma_\tau \sim L_\gamma(\tau) |\tau|^{-\alpha}, \quad |\tau| \rightarrow \infty, \quad 0 < \alpha < 1,$$

with L_γ slowly varying. Equivalently, the spectrum blows up at the origin:

$$\mathcal{S}(f) \sim L_S(f) |f|^{\alpha-1} \xrightarrow{f \rightarrow 0} \infty \quad (\text{a spectral pole}).$$

The full range $0 < \alpha < 2$ gives three regimes: long memory ($0 < \alpha < 1$), short-range dependence ($\alpha = 1$, the ARMA case, $\mathcal{S}(0)$ finite), and anti-persistence ($1 < \alpha < 2$, with $\sum_k \gamma_k = 0$).

Definition — Fractional differencing & FARIMA

The **fractional differencing operator** expands $(1 - B)^d$ for real d :

$$(1 - B)^d = \sum_{j=0}^{\infty} \frac{\Gamma(j-d)}{\Gamma(-d)\Gamma(j+1)} B^j.$$

The pure fractionally differenced process is $(1 - B)^d X_t = \varepsilon_t$; the **FARIMA** class is the stationary solution of

$$(1 - B)^d \phi(B) X_t = \psi(B) \varepsilon_t, \quad d = \frac{1-\alpha}{2} \in (-\frac{1}{2}, \frac{1}{2}),$$

with ε_t i.i.d., ϕ, ψ of degrees p, q with roots outside the unit circle. Integer d gives ordinary ARIMA; $d \in (0, \frac{1}{2})$ gives long memory.

Definition — Self-similarity & fractional Brownian motion

$\{Y_t\}$ is **self-similar** with Hurst exponent H if $Y_{ct} \stackrel{d}{=} c^H Y_t$ for all $c > 0$. With stationary increments this forces

$$\text{Cov}(Y_t, Y_s) = \frac{\sigma^2}{2} (|t|^{2H} + |s|^{2H} - |t-s|^{2H}).$$

The Gaussian process with this covariance is **fractional Brownian motion** $B_H(t)$ (the *only* self-similar Gaussian process); its increments are **fractional Gaussian noise**, with $\gamma_X(\tau) \sim H(2H-1)|\tau|^{2H-2}$. The graph has fractal dimension $D = 2 - H$.

Key results & formulas

Proposition — AR(1) residuals are the noise

For a correctly specified AR(1), $X_t = \phi X_{t-1} + \varepsilon_t$ with known ϕ , the fitted value is $\hat{X}_t = \phi X_{t-1}$, so

$$e_t = X_t - \phi X_{t-1} = \varepsilon_t.$$

Idea: the AR structure cancels exactly, leaving the innovation. This is the template for *all* diagnostics: right model $\Rightarrow$ residuals reproduce the white noise.

Proposition — Box–Pierce & Ljung–Box portmanteau

To test $H_0 : \rho_1 = \dots = \rho_m = 0$ (residuals white up to lag m):

$$\text{Box–Pierce: } Q_m = n \sum_{\tau=1}^m \hat{r}_\tau^2,$$

$$\text{Ljung–Box: } Q_m = n(n+2) \sum_{\tau=1}^m \frac{\hat{r}_\tau^2}{n-\tau}.$$

Both are approximately χ_{m-p-q}^2 for an ARMA(p, q) fit. *Idea:* under H_0 , $\sqrt{n} \hat{r}_\tau \xrightarrow{d} \mathcal{N}(0, 1)$ and asymptotically independent, so a sum of squared normals; fitting $p + q$ parameters costs $p + q$ degrees of freedom. Ljung–Box rescales each term to match the χ^2 better at finite n and is the version used in practice.

Reading the Ljung–Box test

It tests *many* lags at once, dodging the multiple-testing trap of eyeballing the correlogram lag by lag. **Reject** $\Rightarrow$ leftover autocorrelation $\Rightarrow$ model inadequate (go back and re-specify). **Fail to reject** $\Rightarrow$ no evidence against whiteness $\Rightarrow$ model adequate. Watch the df: it is $m - p - q$, **not** m — the noise variance is *not* subtracted, and you need $m > p + q$.

Proposition — the danger of R^2

With $R^2 = 1 - s_e^2/s_y^2$: for an AR(1), $s_y^2 = \sigma^2/(1 - \phi^2)$ and $s_e^2 = \sigma^2$, so

$$R^2 = 1 - \frac{\sigma^2(1 - \phi^2)}{\sigma^2} = \phi^2.$$

Even the *true* model caps at ϕ^2 (e.g. $\phi = 0.4 \Rightarrow R^2 = 0.16$, which "looks bad" but is optimal). Adding parameters can only inflate R^2 . **Use information criteria, not R^2 .**

Proposition — penalty comparisons

Per parameter, AICc charges more than AIC by $\frac{2k(k+1)}{n-k-1} > 0$ (vanishes as $n \rightarrow \infty$, k fixed). BIC charges $\log n$ vs. AIC's 2, so BIC penalizes more iff

$$\log n > 2 \iff n > e^2 \approx 7.39.$$

Hence for any realistic n , BIC is the conservative/consistent choice and AIC the permissive (predictive) one; AICc is the safe default when n is small or k is comparable to n .

Proposition / Theorem — long-memory fingerprints

Variance of the sample mean. Short-range: $\text{Var}(\bar{X}) \sim \mathcal{S}(0)/n$. Under long memory ($\gamma_\tau \sim |\tau|^{-\alpha}$, $0 < \alpha < 1$),

$$\text{Var}(\bar{X}) \sim c n^{-\alpha} \quad (\text{slower than } 1/n),$$

so naive standard errors are far too small. *FARIMA spectrum.* The fractional difference contributes $|1 - e^{-2\pi i f}|^{-2d} \sim |2\pi f|^{-2d}$ near 0, so

$$\mathcal{S}(f) \stackrel{f \rightarrow 0}{\sim} c_f |f|^{-2d}, \quad c_f = \sigma_\varepsilon^2 \frac{|\psi(1)|^2}{|\phi(1)|^2}.$$

The slope of $\log \mathcal{S}$ vs. $\log |f|$ near 0 is $-2d = \alpha - 1$ — the basis of log-periodogram (GPH) estimation of d .

Intuition

Box–Jenkins loop + the ACF/PACF cut-offs

Four iterated steps: **Identify** p, d, q (smallest d that stabilizes the series; read p, q from cut-offs) → **Estimate** → **Diagnose** (residual plots + Ljung–Box; if inadequate, restart) → **Use** (forecast/inference). On the differenced series: $AR(p)$ has ACF *tailing off*, PACF **cutting off after lag p** ; $MA(q)$ is the mirror; ARMA both tail off; very slow decay $\Rightarrow$ re-difference. "Cuts off" = inside $\pm 1.96/\sqrt{n}$ beyond the order.

Where the spectral pole comes from

Differencing $(1 - B)$ has transfer function $1 - e^{-2\pi i f}$, zero at $f = 0$: it kills the trend (high-pass). A *fractional negative power* $(1 - B)^{-|d|}$ inverts a fraction of that, leaving a mild *pole* at $f = 0$. So long memory is "a little bit non-stationary": more low-frequency power than any ARMA, yet still stationary while $d < \frac{1}{2}$.

One phenomenon, three parameters

Long memory is described by α, d , or H , linked by $d = \frac{1-\alpha}{2}$ and $H = d + \frac{1}{2} = 1 - \frac{\alpha}{2}$. Persistent: $0 < \alpha < 1$, $0 < d < \frac{1}{2}$, $\frac{1}{2} < H < 1$, $\mathcal{S}(0) = \infty$. Short-range: $\alpha = 1$, $d = 0$, $H = \frac{1}{2}$ (ARMA / Brownian boundary). Anti-persistent: $1 < \alpha < 2$, $-\frac{1}{2} < d < 0$, $0 < H < \frac{1}{2}$, $\mathcal{S}(0) = 0$.

Ljung–Box test, $n = 150$, ARMA(1,1)

Residual autocorrelations $\hat{r}_{1:4} = 0.03, -0.09, 0.11, -0.05$. With $n(n+2) = 22800$,

$$Q_4 = 22800 \left(\frac{0.03^2}{149} + \frac{0.09^2}{148} + \frac{0.11^2}{147} + \frac{0.05^2}{146} \right) \approx 3.65.$$

Degrees of freedom = $m - p - q = 4 - 1 - 1 = 2$; the 5% critical value is $\chi_{2,0.95}^2 = 5.99$. Since $3.65 < 5.99$ we **do not reject**: no evidence of residual autocorrelation, the ARMA(1,1) is adequate.

Key takeaways

Key takeaways — the week's reflexes

- **Diagnose on residuals**: they should look like the white-noise input — zero mean, constant variance, uncorrelated, Gaussian. Check with time plots, a Q–Q plot, and the residual correlogram ($\pm 1.96/\sqrt{n}$).
- **Run Ljung–Box** $Q_m = n(n+2) \sum_{\tau} \hat{r}_{\tau}^2 / (n - \tau)$ with df = $m - p - q$ (subtract $p + q$, not the variance). Small p -value $\Rightarrow$ refit.
- **Select with information criteria**, never R^2 (which only grows with k): smallest AIC/AICc/BIC wins. AICc for small samples, BIC for parsimony, AIC for prediction.
- **Box–Jenkins loop**: identify (d , then p, q from ACF/PACF) $\rightarrow$ estimate $\rightarrow$ diagnose $\rightarrow$ use.
- **Long memory** = polynomially decaying $\gamma_{\tau} \sim |\tau|^{-\alpha}$ and a spectral pole $\mathcal{S}(f) \sim |f|^{\alpha-1}$; model it with fractional differencing / FARIMA, $d = (1 - \alpha)/2$. Convert freely between α, d, H and read the regime; $\text{Var}(\bar{X}) \sim n^{-\alpha}$ is its empirical signature.